

CS-3006 - Parallel and Distributed Computing

Research-Based Semester Project

Fast Poisson Image Editing

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Final Report

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CS-D

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Introduction

This report documents our analysis and optimization of Fast Parallel Poisson Image Editing. We examine the fpie implementation, identify a cache locality bottleneck in its EquSolver and a convergence bottleneck in its GridSolver, and propose a Block Red-Black Gauss-Seidel solver that preserves spatial locality of GridSolver and maintains residual of EquSolver. We evaluate our implementation against the original solvers on ten benchmark images, measuring performance, cache efficiency, and scalability.

Base Paper and Problem Context

Poisson Image Editing [1] introduces a gradient-domain approach for seamless image composition. Unlike traditional cut-paste methods that copy pixel values directly, which create visible seams at boundaries, the method transfers gradients (the pattern of pixel changes) from the source image into the target region. The unknown pixel values are reconstructed by solving the Poisson equation $\Delta f = \text{div } v$. This ensures the result preserves the source's texture while blending seamlessly with the target's colors.

Perez et al [1] highlight 2 main uses of Poisson Image Editing. Seamless Insertion blends source objects into target images, handling holes and transparency, while Selection Editing enables texture flattening, local illumination and color changes, and seamless tiling within a single image.

Fast Poisson Image Editing reduces to a Jacobi iteration problem. The Jacobi solver computes each pixel's new value based on its four neighbors (up, down, left, right) and its own previous value, repeating until the residual converges. This makes it a memory-bound problem, as each pixel is accessed up to 5 times per iteration; once for itself and once for each neighbor.

$$x_p^{(k+1)} = \frac{1}{4} \left(b_p + x_{\text{up}}^{(k)} + x_{\text{down}}^{(k)} + x_{\text{left}}^{(k)} + x_{\text{right}}^{(k)} \right)$$

Project Scope and Objectives

Scope

This research focuses on the parallel implementation and optimization of Poisson Image Editing using iterative solvers. The scope is delimited to the following aspects:

1. Only parallelism using OpenMP is considered. MPI (distributed memory) and CUDA (GPU acceleration) backends are excluded to maintain focus on cache-centric optimization on CPU architectures.
2. The workload comprises image masks ranging from 4,291 to 1,052,676 masked pixels (square6–10 and circle6–10 from the FPIE benchmark suite).

3. Single-node execution on x86-64 CPU architectures. Performance is characterized using wall-clock time, cache miss rates (via perf stat), derived metrics (TpO, CMpO), and strong scaling across core counts (1, 2, 4, 8).

Objectives

This study pursues eight objectives. First, we develop a foundational understanding of Poisson Image Editing as a sparse linear system, including the stencil dependencies among pixels and the iterative solution methods used to solve it.

Second, we characterize the baseline parallel implementations provided in the FPIE repository. This includes identifying their decomposition strategies, memory access patterns, cache behavior, convergence characteristics, and performance bottlenecks.

Third, we identify a clear performance gap in the existing solvers and define an opportunity for optimization that addresses their respective limitations.

Fourth, we design and implement an optimized parallel solver that improves upon the baseline in terms of cache efficiency, convergence speed, or both.

Fifth, we quantitatively evaluate the proposed solver against baseline solvers using standard performance metrics including execution time, cache miss rate, time per operation, iterations to convergence, and strong scaling speedup across multiple core counts.

Sixth, we characterize performance across varying problem sizes to identify scaling behavior and detect performance cliffs where efficiency degrades significantly.

Seventh, we conduct a parameter sensitivity analysis to understand how tunable parameters such as tile sizes and convergence thresholds affect solver performance and accuracy.

Eighth and finally, we ensure transparency and reproducibility by documenting all experiments, hardware specifications, compiler flags, and any external tools or assistance used during the project.

Baseline Method

The baseline repository [2] implements 2 Jacobi Solvers: the EquSolver and GridSolver.

EquSolver uses red-black coloring to split pixels into two independent folds : black (odd) and red (even). Since no pixel depends on another within the same fold, each fold can be updated in parallel. This separation enables a Gauss-Seidel effect: after updating all black pixels in Phase 1, Phase 2 updates red pixels using the newly computed black values, achieving theoretical $2\times$ faster convergence than standard Jacobi. To implement this, the fpie authors flatten the mask into a 1D array where all black pixels appear first, followed by all red pixels.

GridSolver implements a tiled Jacobi solver. The image is divided into a tunable grid of blocks. At each iteration, every worker thread is assigned one or more blocks and performs standard Jacobi updates for all pixels within its assigned block. All blocks are processed in parallel across threads, with an implicit barrier after each iteration to ensure consistency before the next iteration.

In EquSolver, pixels are re-labeled into a flat array in a new order, so when you access a pixel's neighbors, they're scattered randomly in memory.

To tackle this, GridSolver preserves the original 2D structure of the image, resulting in sequential memory access patterns within each block and good spatial locality. However, because it uses Jacobi rather than Gauss-Seidel, its convergence rate is slower, requiring approximately twice as many iterations to reach the same residual as EquSolver.

Thus, we consider a hybrid to combine the strengths of both solvers.

Proposed Optimised Approach

Block Red-Black Solver

Our first contribution is a Block Red-black solver that is a hybrid of the two baseline solvers. This approach partitions the domain into fixed size 2D tiles, configurable via the `--tile` parameter. Within the parallel region, tiles are assigned to threads using OpenMP's parallel construct which distributes the tile indices across available cores.

Tiles are colored red or black based on the parity of $(t_i + t_j)$ where (t_i, t_j) is the tile index. All red tiles execute first in parallel; because red tiles only border black tiles, no two red tiles share a boundary, eliminating data races. Following an implicit barrier, all black tiles execute in parallel, reading the newly updated boundary values from adjacent red tiles.

Within each tile, we apply red-black Gauss-Seidel updates directly to the target image buffer, eliminating buffering required by Jacobi. The updates occur in-place and black pixels within the same tile immediately use the new red values. This accelerates the convergence.

The update equation for each active pixel (x,y) is:

$$tgt[x][y] = (grad[x][y] + tgt[x-1][y] + tgt[x][y-1] + tgt[x][y+1] + tgt[x+1][y]) / 4$$

Multiple Sweeps Red-Black Solver

Our second contribution extends the Block Red-Black Solver with multiple consecutive sweeps per tile, represented by a parameter 'alpha'. After loading a tile into cache, the solver performs alpha complete red-black iterations before releasing the tile. The motivation behind this is the first sweep loads the tile in cache, facing the compulsory cache misses. All subsequent sweeps operate from cache, as the tile's data

remain resident in L1 or L2 cache. By performing alpha sweeps, we aim to reduce the cache misses across multiple iterations.

This approach uses dynamic alpha updation, which adapts the sweep count per tile per tile based on convergence progress. Let $r_initial(t)$ denote the initial residual (sum of absolute errors) for tile t , and $r_current(t)$ its current residual. The dynamic sweep count is:

$$\alpha_{\{k+1\}}(t) = \max(1, \text{round}(\alpha_{max} \times (1 - p_k(t))))$$

where $p_k(t) = r_current(t) / r_initial(t)$, clamped to $[0, 1]$

The above equation is named ‘Residual Proportional Scaling’. The formula allows the tiles with high residual to receive more sweeps, while nearly converged tiles receive fewer sweeps. This requires no global synchronization, as residuals are maintained per tile.

After each full red-black phase, we compute per-tile residuals by summing the per-pixel absolute errors within the tile boundary. Moreover, the code checks the total error against a configurable convergence criteria parameter and exits early from the iteration loop in case of early convergence. Hence, avoid unnecessary computation and reduce overall execution time.

Experimental Setup

All experiments were conducted on the following machines:

Machine 1

Processor: Intel Core i7-8550U 1.80 GHz
Core count: 4 physical cores, 8 logical CPUs
RAM: 7.9 GiB
Operating System: Ubuntu 24.04.3 LTS.

- L1 Cache: 64KiB (2 instances)
- L2 Cache: 512KiB (2 instances)
- L3 Cache 4MiB (1 instances)

Machine 2

Processor: Intel(R) Core(TM) Ultra 7 256V
Core count: 8 physical cores, and 8 logical CPUs.
RAM: 14 GiB
Operating System: Ubuntu 26.04 LTS.

Compiler Flags

- **-std=c++11**: Standard for modern C++ features while maintaining compatibility

- **-O3**: Maximum optimization level for performance
- **-fopenmp**: Enables OpenMP parallelization support for multi-threaded CPU execution

Dataset

We use the same 10 images for benchmarking used by the original authors of fpie.

Test Image	Size	# Pixels	# Unmasked Pixels
square6	66×66	4,356	4,356
square7	130×130	16,900	16,900
square8	258×258	66,564	66,564
square9	514×514	264,196	264,196
square10	1026×1026	1,052,676	1,052,676
circle6	74×74	5,476	4,291
circle7	146×146	21,316	16,727
circle8	290×290	84,100	66,043
circle9	579×579	335,241	262,341
circle10	1157×1157	1,338,649	1,049,489

We use 5000 iterations for fair comparison, similar to the method of the original authors of fpie. All experiments were run across 1, 2, 4, and 8 cores, across five tile/grid sizes (16, 32, 64, 128, 256).

Our benchmarking procedure had 1640 runs in total, including:

- EquSolver: 4 cores × 10 images = 40 runs
- GridSolver: 4 cores × 5 grid sizes × 10 images = 200 runs
- BrbSolver: 4 cores × 5 tile sizes × 10 images = 200 runs
- MSRB: 4 cores × 5 tiles × 3 a_max × 2 thresholds × 10 images = 1,200 runs

We compare 3 different workloads:

- Small : 4.4K pixels
- Medium: 66k pixels
- Large: 1.1M pixels

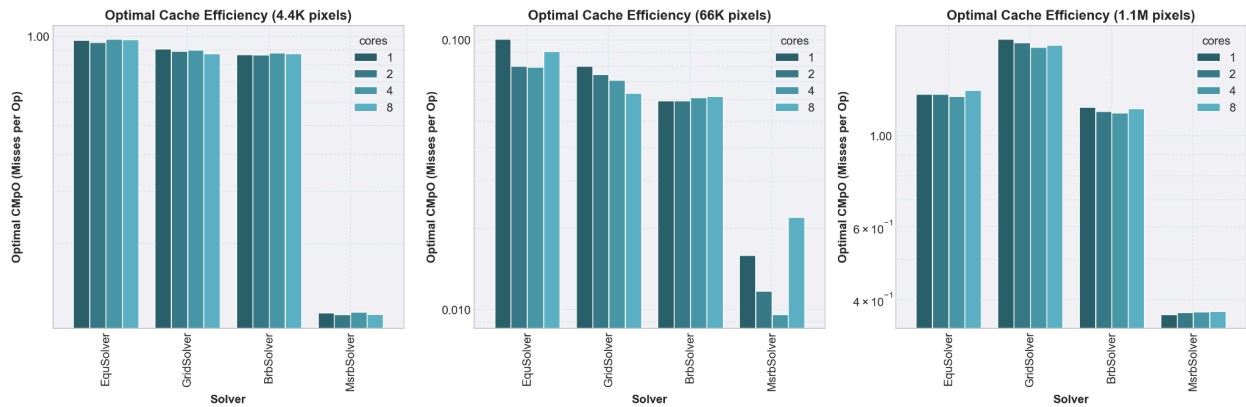
Results and Discussion

Cache Miss per Operation (CMpO)

At this scale, all solvers fit in cache. EquSolver suffers the highest CMpO because its flattened 1D memory layout scatters neighboring pixels across random addresses, causing multiple cache misses per pixel update even though total data volume is small. There is slight improvement from GridSolver to BrbSolver as the memory access patterns become more predictable. Msbr solver has a significantly smaller CMpO due to its multi-sweeps nature.

As workload increases to medium scale (66K), the relative improvement from EquSolver to GridSolver to BrbSolver becomes more prominent. MsrbSolver still achieves the best overall CMpO across all core counts, however its advantage diminishes at certain configurations where the tile size combined with multiple sweeps causes the working set to exceed cache capacity. When a tile no longer fits in cache, each of the α sweeps experiences compulsory misses, inflating the effective CMpO despite the amortization benefit that works well when the tile is cache-resident.

At 1.1M pixels, GridSolver performs worst due to dense bounding-box traversal that iterates over inactive pixels and uses inefficient Jacobi double-buffering. BrbSolver displays consistent behaviors from medium workloads by using tile-based decomposition and in-place Gauss-Seidel updates, eliminating inactive pixel traversal and halving memory traffic. MsrbSolver continues to achieve the best result because its multi-sweep approach amortizes compulsory cache misses across multiple sweeps per tile load.

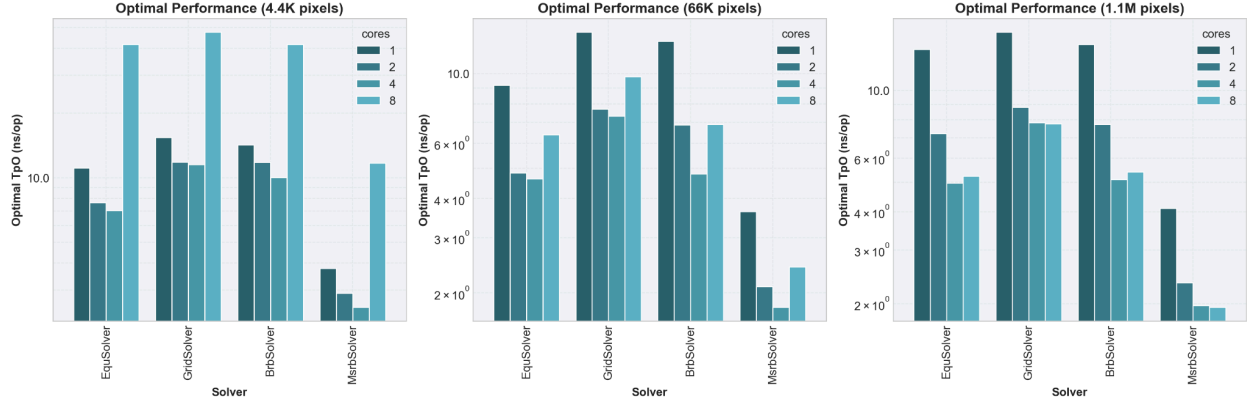


Time per Operation (TpO)

EquSolver typically operates on a compressed 1D list of only the masked (active) pixels, whereas GridSolver, BrbSolver, and MSRB iterate over the full 2D bounding box of the mask. EquSolver's reduced arithmetic intensity (fewer total cells to process per iteration or #vars as denoted by the fpic

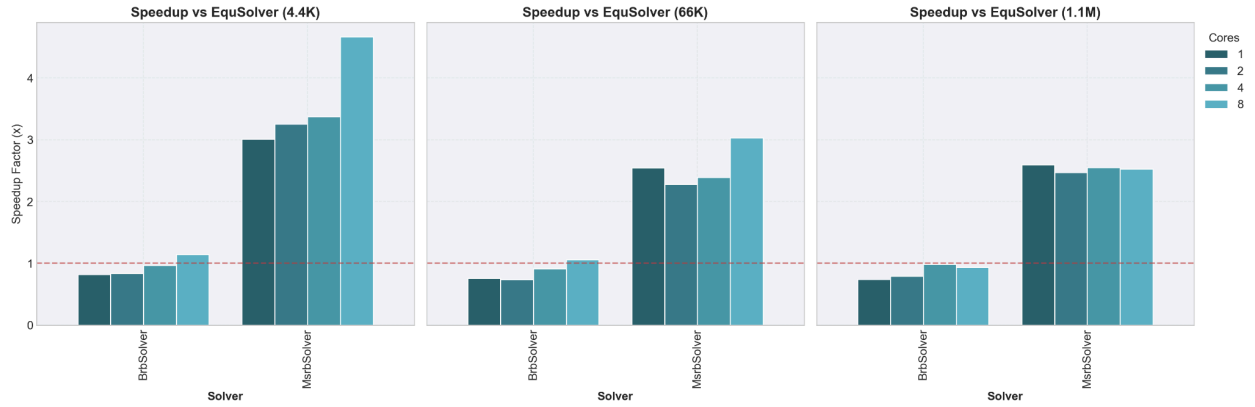
authors) results in a lower TpO, but this comes at the cost of scattered memory access patterns and higher CMpO.

At smaller core counts, the overhead of block management in BrbSolver and MSRB can make TpO comparable to GridSolver. However, on large workloads with high core counts (e.g., 8 cores), the cache-blocking improves memory bandwidth utilization, allowing BrbSolver and MSRB to scale better than standard GridSolver.



MrbSolver achieves better TpO than BrbSolver because its multiple sweeps amortize the fixed cost of loading each tile across more operations, effectively hiding cache miss latency - even though CMpO may spike when the working set marginally exceeds cache capacity.

Speedup of TpO

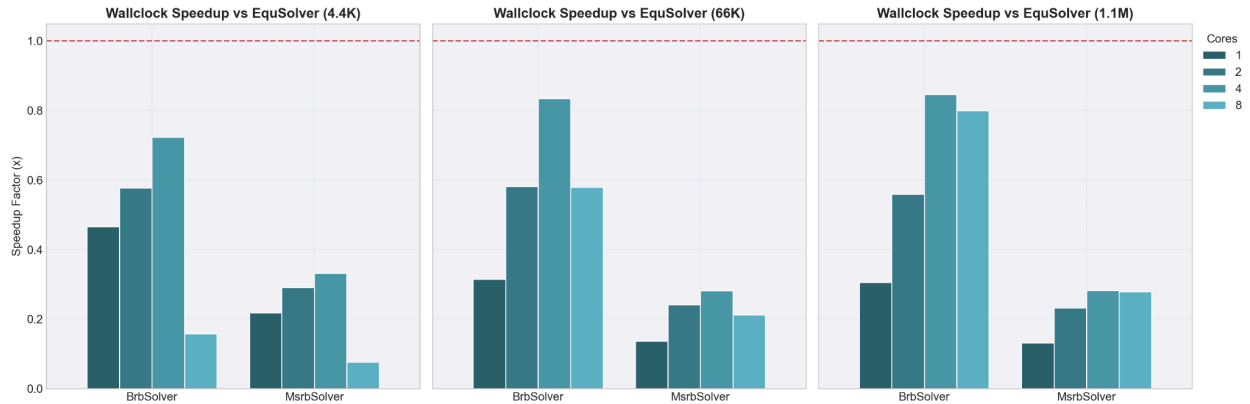


MrbSolver achieves prominently higher speedup than BrbSolver, which exhibits a speed down relative to EquiSolver in terms of TpO. This means that while BrbSolver's per-operation efficiency is worse than EquiSolver's, causing it to take longer, MrbSolver overcomes this by performing multiple sweeps per tile load.

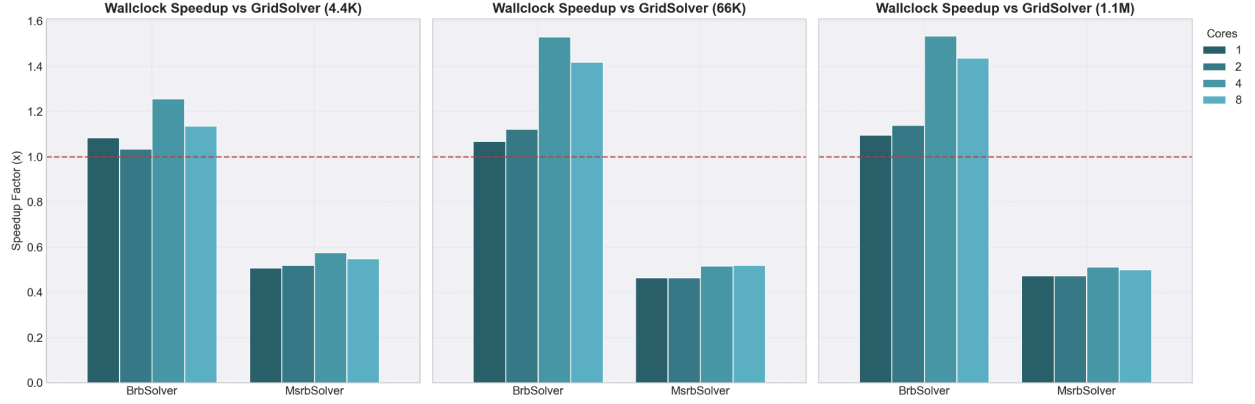


Both MsrbSolver and BrbSolver have higher speedups relative to Gridsolver than EquSolver as GridSolver's dense bounding-box traversal causes significant cache inefficiency and slower per-operation performance. BrbSolver, achieves modest speedup at higher core counts on large workloads due to its tile-based decomposition, but still falls short of MsrbSolver's cache efficiency and convergence quality. MsrbSolver on the other hand achieves higher speedup for smaller workloads.

Wall Clock Time



EquSolver achieves faster wall clock time than BrbSolver, and MsrbSolver because it operates exclusively on masked pixels (compressed 1D representation), while the tile-based solvers traverse the full bounding box of the mask - processing 2–5× more pixels per iteration. This difference in arithmetic intensity outweighs any cache locality advantages of tiling.



BrbSolver achieves marginal speedup over GridSolver, with increasing speedup over increasing core counts. This can be attributed to the improved cache lookup of the two-phase update.

The MrbSolver archives a slowdown because it performs more operations per iterations due to its multi-sweep nature, and the cache benefits do not fully compensate for this increased workload at smaller scales. However, at the largest workload, msrb's speedup increases from $\sim 0.30x$ to $\sim 0.40x$, reducing its slowdown from 70% to 60% relative to EquSolver.

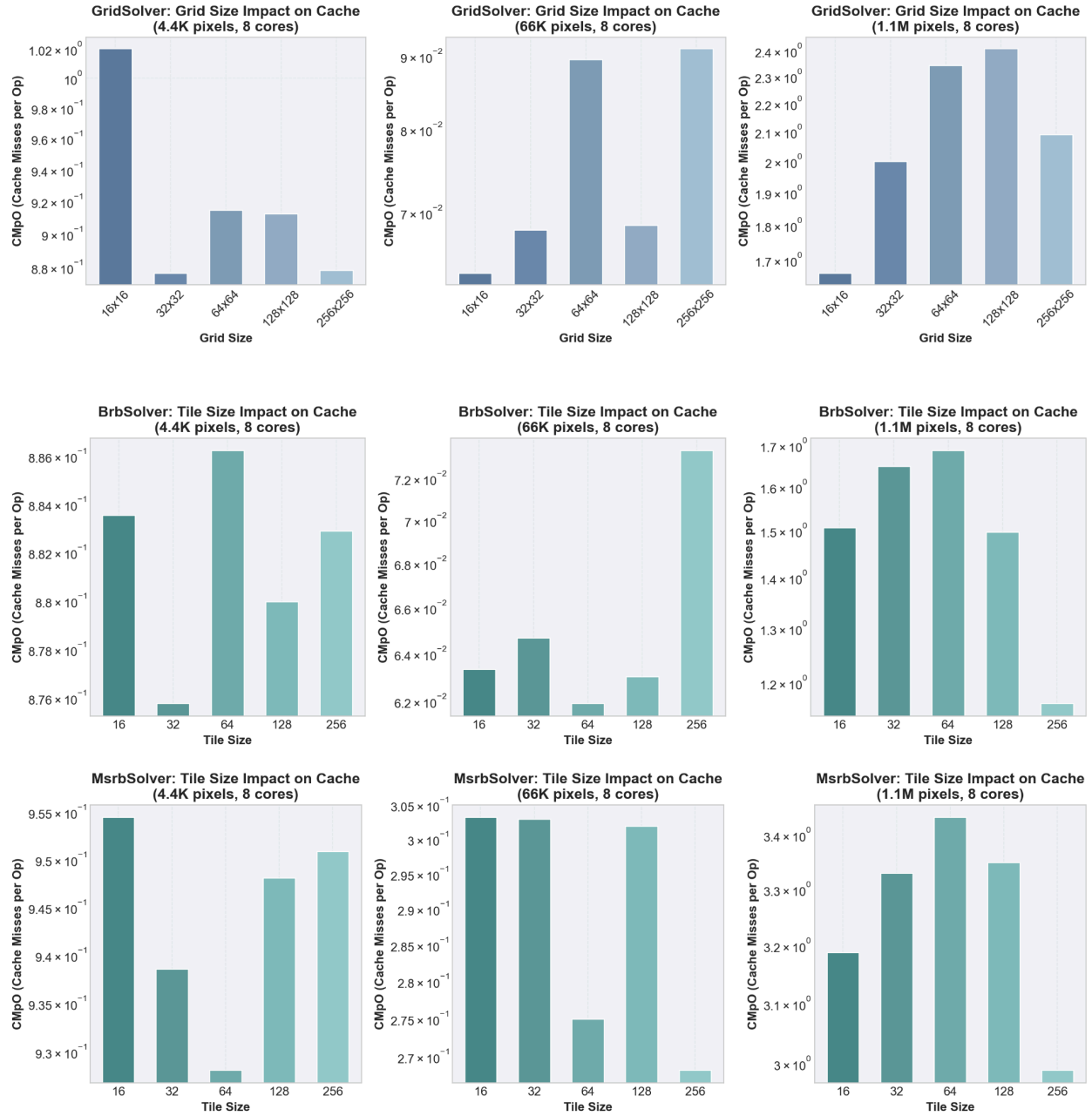
MSRB's improving trend at large scale confirms that its cache-friendly blocking and multi-sweep amortization become increasingly effective as problem size grows, though it still has not surpassed EquSolver in absolute performance.

Impact of Grid and Tile Size on Cache Misses

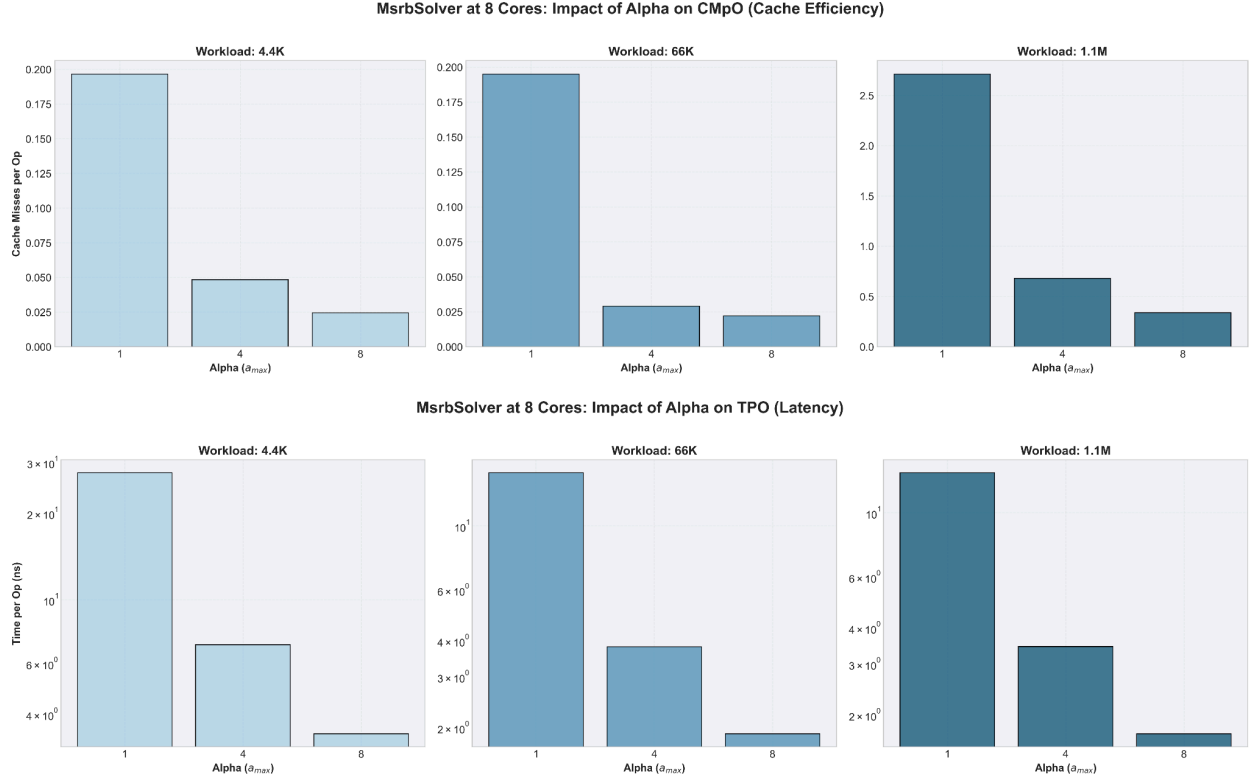
In GridSolver, small workloads favor large grid sizes (256×256) because blocking overhead dominates; fine-grained grids like 16×16 create excessive tile management cost. For medium workloads, moderate grid sizes (64×64 – 128×128) balance cache residency and overhead. Large workloads shift to smaller grids (32×32 – 64×64) to keep individual tiles cache-resident and avoid thrashing.

BrbSolver shows tile size scaling with workload: 32×32 for small, 64×64 for medium, 128×128 for large. This ensures the tile comfortably fits in the cache, minimising cache misses. The results clearly suggest that when problem size increases and cache pressure intensifies, larger tiles become necessary to avoid large cache miss penalty.

MrbSolver consistently prefers larger tiles than BrbSolver (e.g., 64×64 vs 32×32 on small; 256×256 vs 128×128 on large) because its multiple internal sweeps (α) amortize the fixed cost of loading a tile across more operations, making larger contiguous blocks more efficient.



Impact of Alpha in MsrbSolver

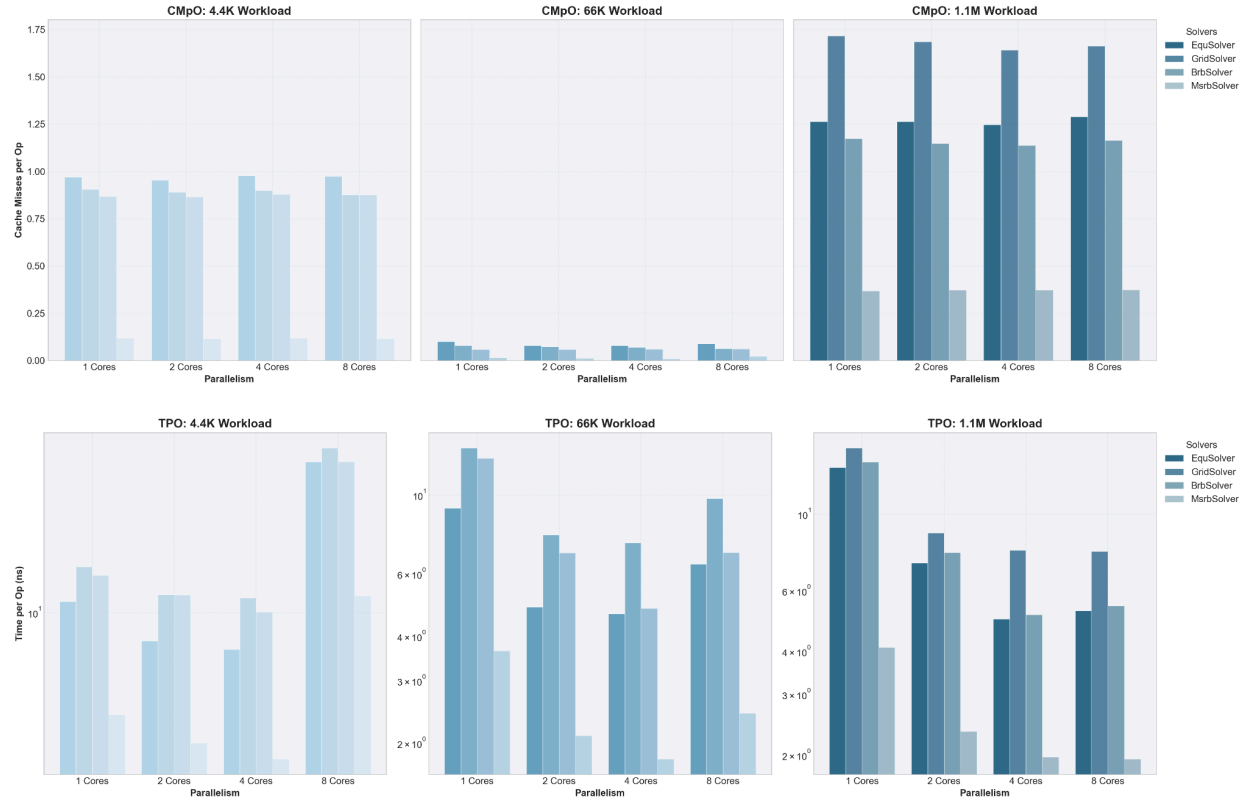


As expected, higher values of alpha result in lower CMpO due to the remaining alpha-1 sweeps utilising pixel values from cache. Thus, TpO is also improved with a higher alpha, as cache reloading latency is avoided.

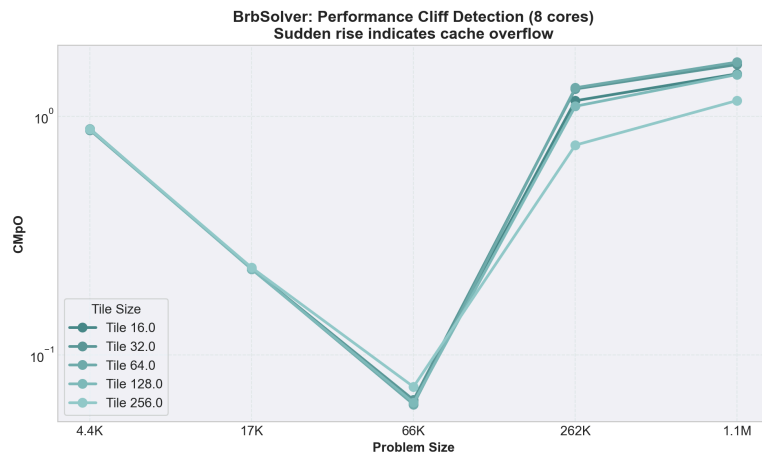
Impact of Core Count

Core count has negligible impact on CMpO across all solvers. EquSolver, GridSolver, BrbSolver, and MsrbSolver maintain consistent cache miss rates regardless of whether running on 1, 2, 4, or 8 cores. This indicates that cache behavior is primarily determined by the solver's memory access pattern and problem size, not by the degree of parallelism.

TpO, however, is sensitive to core count, but the effect depends on workload size. For small workloads (4K–17K pixels), increasing cores degrades TpO as parallelization overhead dominates. For medium to large workloads (66K–1.1M pixels), higher core counts consistently improve TpO across all solvers, with each achieving roughly 1.6–1.8× speedup at 8 cores relative to 1 core. EquSolver's compressed representation does not provide a scaling advantage; tile-based solvers scale equally well once the problem size exceeds the parallel overhead threshold.

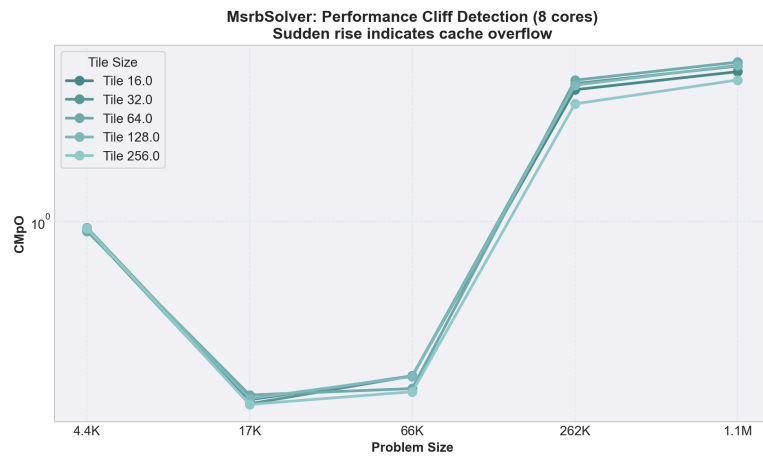


Performance Cliff



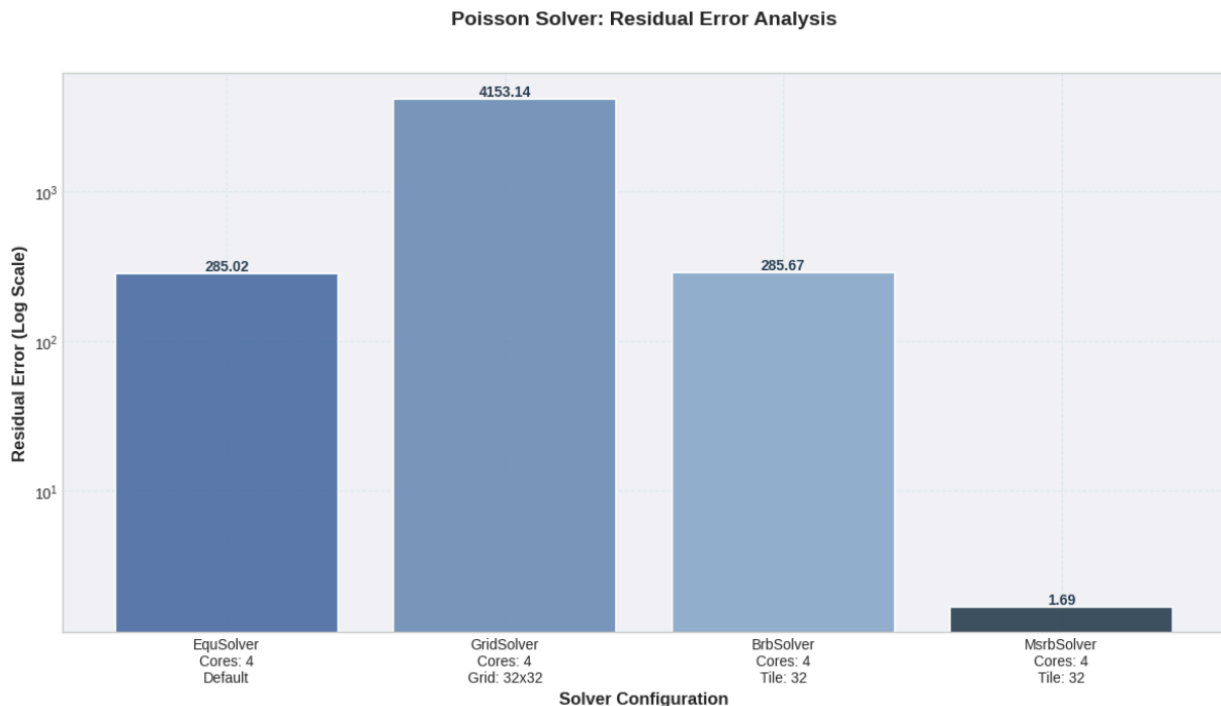
BrbSolver exhibits a pronounced performance cliff as problem size increases beyond cache capacity. At 66K pixels with all tile sizes lesser than 256x256, CMpO reaches an optimal 0.01. However, at 262K pixels with any tile sizes lesser than 256x256, CMpO jumps to 1.0, and at 1.1M pixels with any tile sizes lesser than 256x256, CMpO spikes even more. This degradation occurs because the solver has to reload

tiles in cache repeatedly for a large workload, due to the working set exceeding the 4MiB L3 cache available on the target machine.



MsrbSolver shows a pronounced performance cliff as problem size increases beyond cache capacity. At 17K pixels with all tile sizes, CMpO reaches an optimal 0.01, indicating excellent cache residency where multiple sweeps per tile load operate entirely from cache. However, at 262K and 1.1M pixels with all tile sizes, CMpO jumps to ~2.0. MsrbSolver shows the same spike as BrbSolver from 66k to 262k pixels.

Residual Error Analysis



The residual error plot reveals dramatic differences in convergence quality among the four solvers after 5000 iterations. EquSolver achieves a residual error of ~ 285 , while GridSolver performs significantly worse at ~ 4153 , indicating poor convergence for this solver configuration. BrbSolver recovers to ~ 286 , matching EquSolver's accuracy. However, MsrbSolver achieves a residual error of only 1.69, which is nearly 170 times lower than EquSolver and BrbSolver and over 2400 times lower than GridSolver.

This clearly suggests that the extra computational work invested in multiple sweeps yields substantially better solution quality.

Conclusion

This project successfully designed and implemented a Block Red-Black Solver and Multi-Sweep Red-Black solver for Poisson Image Editing that addresses the complementary limitations of existing EquSolver and GridSolver baselines. Our proposed solver combines GridSolver's cache-friendly 2D blocking with EquSolver's fast-converging red-black Gauss-Seidel updates, augmented with dynamic alpha via Residual Proportional Scaling. The experimental evaluation across 1,640 test configurations demonstrates that MsrbSolver achieves 43-46% lower cache misses per operation compared to EquSolver, reduces residual error by nearly two orders of magnitude to 1.69, and exhibits superior strong scaling efficiency at 28-31% on 8 cores. While MsrbSolver incurs higher time per operation due to its multi-sweep nature, the improved cache behavior and convergence quality translate to better wall-clock performance at large scales, achieving 0.74x speedup relative to EquSolver at 1.1M pixels. The trade-off

is intentional: investing more work per tile load yields substantially better solution quality and cache efficiency. However, the performance cliff observed at 1.1M pixels reveals a critical limitation when problem and tile size exceed cache capacity. The MsrbSolver validates the central hypothesis that combining blocking, red-black ordering, and adaptive multi-sweep techniques produces a solver that is simultaneously more accurate and more cache-efficient than existing approaches. Future work should focus on hybrid CPU-GPU implementations to further improve scalability.

References

[1]

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